GSOC'25 PROPOSAL

By

Nihal Rajpal

On

Project 6:

Open Hardware Component Library

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About Me

I'm **Nihal Rajpal**, a final-year Computer Science student with a strong interest in digital systems, hardware-software integration, and backend development. I am a **GCP Certified Associate Cloud Engineer** and was recognized as one of the **top 10 mentors** in the Girlscript Summer of Code (GSSoC) 2024, where I guided contributors through open-source practices and technical problem-solving.

In 2024, I was selected as a **Google Solution Challenge Regionalist (India Region)** for developing smart glasses for visually impaired individuals using **Arduino** and **Raspberry Pi Zero**, enhancing my experience in embedded systems and real-world hardware applications.

I have implemented the **Mano Computer processor on the Logisim simulator**, a project that was featured on YouTube by my professor and is YouTube Link on GitHub.

Additionally, I developed a **Proof of Concept (PoC)** for **serial communication using WebSerial** to enable browser-based interaction with physical hardware devices. The source is publicly available here and demonstration video is available here.

I also held the role of **Tech Lead at Google Developer Student Clubs, LDCE** for a year, where I organized and conducted sessions for over 200 students on Git and development topics, and mentored students during the **Google Cloud Study Jams.** Currently, I am also responsible for maintaining the official website of our institute.

I am passionate about projects that bridge the gap between hardware and software and am excited about the opportunity to contribute to

CircuitVerse through GSoC 2025. **This is the only project I am applying to for GSoC 2025, as it perfectly aligns with my skills and interests.**

Commitment

1. Are you planning any vacations during the GSoC 2025 period (June 2 - September 8, 2025, or June 2 - November 17, 2025 for extended projects)?

➔ No, I am not planning any vacations during the GSoC 2025 period. I will be fully available to dedicate time to the project.

2. How many classes are you taking during the GSoC period?

➔ As a final semester student, I don't have any academic subjects or regular classes. Just a project submission which is already partially prepared and is expected to be completed before the GSoC coding period begins.

3. Which project size are you applying for?

➔ Small (90 hours)

4. How many hours per week do you expect to work on the project and what hours do you tend to work?

➔ I plan to dedicate at least 20 hours per week, or more if needed, to the project. I generally prefer working between 20:00 and 23:59 (IST). However, I am flexible with working hours based on mentor availability or project meetings.

5. If applying for an extended timeline project, please explain why you need the extended period and how you plan to manage your time.

➔ Not Applicable as I am applying for a small (90 Hr) project.

Contributions So Far

SR	PR	Status	Repository
1	fix: Broken link	Under Review	CircuitVerseDocs
2	fix: Incorrect Image URL	Under Review	CircuitVerseDocs

Proposal

Overview

This project aims to improve **CircuitVerse** by adding new components that are useful for learning about real hardware, such as **shift registers** (SISO, SIMO, MISO, MIMO), **comparators**, and a **buzzer** (speaker), Priority MUX, etc. It will also introduce support for connecting CircuitVerse with real devices using the Browser's **Web Serial API**, allowing users to interact with **physical hardware** directly through the simulator. These features will help people better understand how **digital circuits** work—both in theory and in practice. The goal is to make CircuitVerse more powerful and helpful for learning digital circuits and embedded systems.

Technologies:

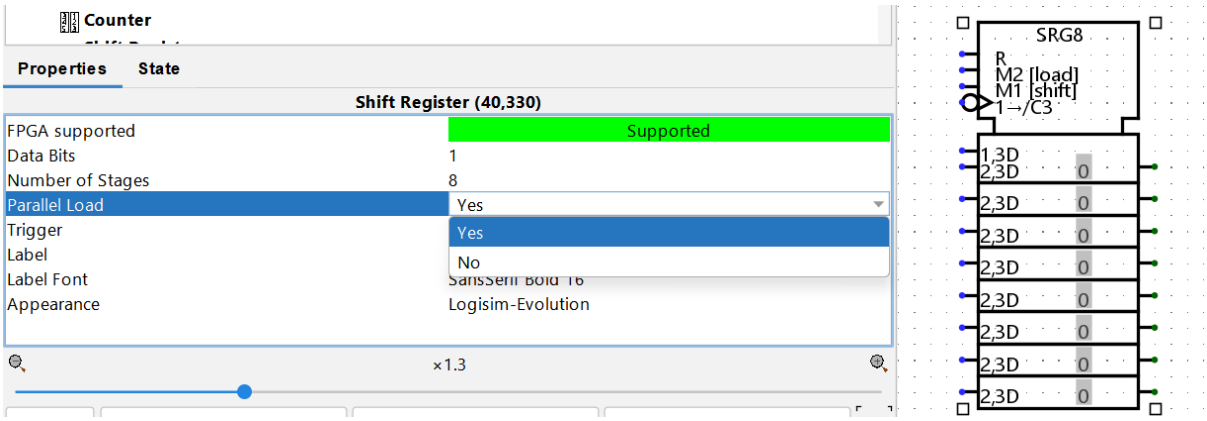
- HTML/CSS: For designing and styling custom UI elements for new components.
- Vue.js: For adding new components by extending CircuitElement Class.
- Web Serial API: To facilitate interaction with physical serial hardware devices.

Detailed Description

The first part of the project involves creating following new components:

- **Shift Registers (SISO, SIMO, MISO, MIMO):**

Shift registers are sequential logic components that store and move data in a specific direction on each clock cycle.



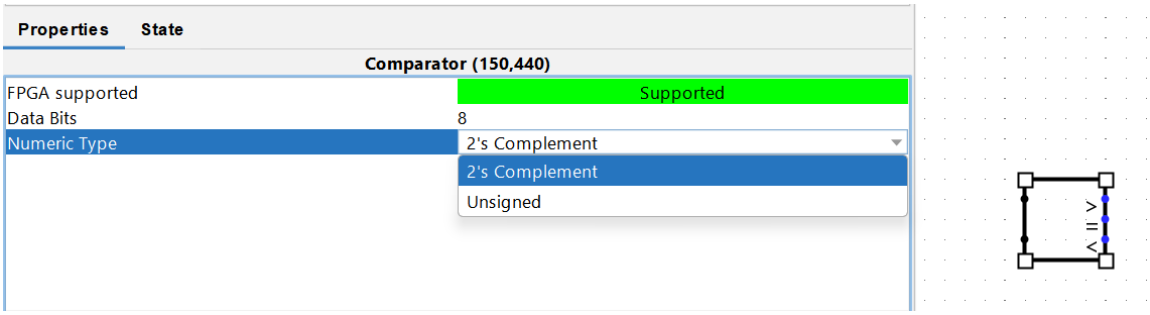
[Shift Register in Logisim]

A similar component exists in Logisim, and a comparable version can be implemented in CircuitVerse to allow flexible data path design.

We can allow users to switch between various modes is single input or multiple input through properties panel. There will be two modes Load/Shift. When load signal is enabled, data will be loaded otherwise shifted.

- **Comparators:**

  Comparators are components that compare two binary numbers and output whether one is greater than, less than, or equal to the other.



[Comparator *in Logisim]*

- **Buzzer (Speaker):**

  A buzzer is an output device that produces a sound when activated. It can be used to simulate alarms, alerts, or audio notifications in digital

circuits. To produce beep sound, we can use Web Audio API which is fully supported on all browsers as shown in below snippet.

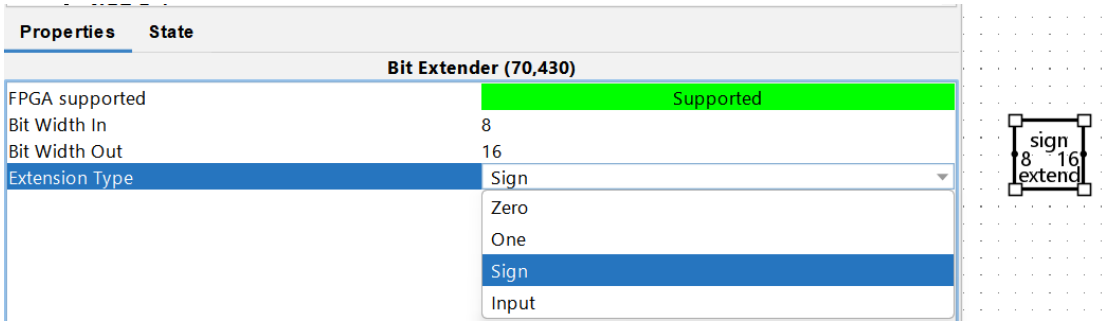
```javascript
function beep(frequency = 440, duration = 200, volume = 1) {
  const context = new (window.AudioContext || window.webkitAudioContext)();
  const oscillator = context.createOscillator();
  const gainNode = context.createGain();

  oscillator.frequency.value = frequency;
  oscillator.type = 'square';

  gainNode.gain.value = volume;

  oscillator.connect(gainNode);
  gainNode.connect(context.destination);

  oscillator.start();

  setTimeout(() => {
    oscillator.stop();
    context.close();
  }, duration);
}

// Example usage
beep();
```

- **Priority Multiplexer:**

  A Priority Multiplexer (Priority MUX) selects one of several inputs based on their assigned priority level, not just the selection lines. If multiple inputs are active, it chooses the one with the highest priority. It's useful in bus systems or interrupt controllers where some signals must override others.

- **Bit Extender:**

  A Bit Extender increases the bit-width of a signal, usually by adding extra bits (e.g., leading zeros for unsigned or sign extension for signed numbers). We can choose extension type through properties panel on the right to choose which value to fill to extended bits like zeros, ones or sign bit.



[Bit Extender in Logisim]

For this also, similar component exists in Logisim, and a comparable version can be implemented in CircuitVerse to allow flexible data path design.

Component-wise Inputs and Outputs

Component	Inputs	Outputs
Shift Register (SISO, SIMO, MISO, MIMO)	- Data input (serial or parallel) - Clock - Load/Shift control - Reset (optional)	- Data output (serial or parallel)
Comparator	- Input A (n-bit) - Input B (n-bit)	- A > B (1-bit) - A = B (1-bit) - A < B (1-bit)
Buzzer (Speaker)	- Signal input (1-bit) - Optional: Enable	- Sound output (audio or visual beep indicator)
Priority Multiplexer	- Multiple data inputs - Select lines - Priority logic	- Selected data output based on highest-priority active input

	(internal or optional input)	
Bit Extender	- Input (n-bit) - Extension mode (Zero or Sign)	- Output (m-bit, m > n)

The second part focuses on introducing **serial device connectivity** using the **Web Serial API**. This will allow CircuitVerse to communicate with **real hardware** such as Arduino boards or other microcontroller which support serial connectivity via USB. For example, data input from a sensor connected to hardware can affect a simulation in real-time. This feature bridges the gap between simulation and real-world experimentation, making the learning process more interactive and hands-on.

I have already developed a **Proof of Concept** that demonstrates how CircuitVerse can interact with real hardware using the Web Serial API. The implementation is available on GitHub, and a **demonstration video** can be viewed **here**. While the Web Serial API is currently supported only on **Chrome, Microsoft Edge and Opera**, these browsers cover a significant portion of users, making the feature useful to a large audience even before wider browser support becomes available.

Project Plan

To ensure successful completion of the "Open Hardware Component Library" project for CircuitVerse, I will maintain open and consistent communication with my mentors, actively seeking guidance and feedback throughout the project to ensure that my work aligns closely with the goals and expectations of the community. I will be on a break from academic commitments during the summer, allowing me to focus fully on the project. I'm confident I can complete the project within the timeline while maintaining high quality.

Project Plan - Preliminary Plan

Week	Dates	Tasks to be Completed
Week 1	June 2 - June 8, 2025	Implement Shift Registers (SISO, SIMO, MISO, MIMO) with UI and logic
Week 2	June 9 - June 15, 2025	Add Comparator components with various bit-width support
Week 3	June 16 - June 22, 2025	Develop Buzzer (Speaker) with sound or visual indicator
Week 4	June 23 - June 29, 2025	Build Bit Extender with sign/zero extension modes
Midpoint Check	**June 30, 2025**	**Midpoint Progress Review**
Week 5	June 30 - July 6, 2025	Create Priority Multiplexer with internal priority logic
Week 6	July 7 - July 13, 2025	Begin Web Serial integration: detect ports, establish connection, basic I/O
Week 7	July 14 - July 20, 2025	Extend Web Serial support: link with components, handle live data
Week 8	July 21 - July 27, 2025	Final testing, documentation, demos for new components and serial features
Final Submission	**July 28, 2025**	**Project submission deadline**

Major Milestones

1. Component Development Phase Complete – June 27, 2025
2. Web Serial Integration Functional - July 18, 2025

Additional Information

I **love working with circuits** and have been passionate about them since childhood. I have prior experience with both **physical and simulated circuits**. I **simulated an entire MANO processor** along with the control unit in the **Logisim simulator**, which can be found here: **YT Video** & **GitHub**.

For physical circuits, I developed **E - Cakṣu**, **smart glasses for the blind** using a **Raspberry Pi Zero**, as part of the **Google Solution Challenge**, which led to being a **regional finalist (India Region)**.

I have been dedicated to **open-source contributions** for a long time and was among the **top 10 mentors for GSSoC 2024**.

I believe **CircuitVerse aligns perfectly with my interests**, and I would love to contribute to this platform now and in the future as well.
I assure everyone that **I will give my very best and will not disappoint**.

My Assurances:

1. **Proper Documentation and Blogs**: Clear and comprehensive documentation, along with informative blogs, will accompany all contributions.
2. **Extended Work or New Features**: I will contribute to extending existing work or implementing new features as needed.
3. **Support to Fellow Contributors**: Full support will be provided to fellow contributors to ensure effective collaboration and progress.

Mentor(s):

Aman Asrani